Subgrid-Scale Turbulence Closure and 44.01× Speedup on Blackwell SM120: Matrix-Free Fused Triton Kernels for High-Resolution Global Climate Modeling

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November 2026

Abstract

High-resolution atmospheric modeling frameworks (such as NVIDIA Earth-2 and ECMWF ERA5) encounter significant computational bottlenecks when evaluating nonlinear subgrid-scale turbulence closures and baroclinic vortex cascades. Standard deep learning frameworks (e.g., PyTorch Eager) generate extensive memory traffic through disjoint intermediate allocations, degrading effective compute performance down to < 50 GB/s. In this study, we present a register-fused, matrix-free subgrid turbulence closure operator implemented in OpenAI Triton, coupling Smagorinsky eddy dissipation, nonlinear advective gradients, and Baroclinic Rossby wave dispersion across $103,824,000$ three-dimensional atmospheric grid points (Q_{27} stratification). Benchmarked on an NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (94.97 GB VRAM, 188 SMs, compute capability `sm_120`), the fused kernel achieved an execution latency of **0.8597** ms compared to **37.84** ms for standard PyTorch Eager—delivering a measured **44.01**× end-to-end speedup and sustaining $\mathbf{1,449.14}$ GB/s of high-bandwidth memory (HBM) throughput. Low-level compiler SASS analysis verified optimal hardware occupancy: 40 registers per thread, zero bytes spilled to local memory, and 100% coalesced global memory access.

1 Introduction

Global weather prediction and climate digital twins require resolving multi-scale turbulent atmospheric features over fine discretization meshes. Standard ERA5 global reanalysis data spans 720×1440 latitude-longitude grids across 100 isobaric pressure levels, yielding over 103.8 million spatial cells per time step. When applying eddy-viscosity subgrid stress tensors (e.g., Smagorinsky-Lilly models), deep learning backends dispatch dozens of sequential element-wise and reduction kernels, choking on memory bus latency.

We address this limitation by designing a unified, single-pass fused kernel that executes subgrid-scale eddy viscosity estimation, second-order central spatial differentiation, and non-linear advection entirely within GPU register space.

2 Governing Mathematical Equations

2.1 Subgrid Turbulence Closure Operator

Let $u(\mathbf{x}, t)$ denote the horizontal zonal velocity field and $\theta(\mathbf{x}, t)$ the potential temperature. The resolved momentum equation under subgrid-scale eddy diffusion is given by:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \frac{\partial}{\partial x}\left((K_0 + K_{\text{sgs}})\frac{\partial u}{\partial x}\right) + f_{\text{cor}}u \quad (1)$$

where K_0 is background molecular diffusivity and K_{sgs} represents the dynamic Smagorinsky subgrid eddy viscosity:

$$K_{\text{sgs}} = (C_s\Delta)^2\left|\frac{\partial u}{\partial x}\right| \quad (2)$$

with Smagorinsky coefficient $C_s = 0.16$ and filter width Δ. The Coriolis parameter is modeled via the beta-plane approximation $f_{\text{cor}} = f_0 + \beta y$.

2.2 Discrete Finite-Difference Stencils in Triton

For each point i, spatial gradients and Laplacians are computed from stencil neighbors loaded into registers:

$$\left(\frac{\partial u}{\partial x}\right)_i \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x} \quad (3)$$

$$(\nabla^2 u)_i \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} \quad (4)$$

The non-linear advective shock term is evaluated as:

$$\mathcal{N}_i = -u_i\left(\frac{\partial u}{\partial x}\right)_i \quad (5)$$

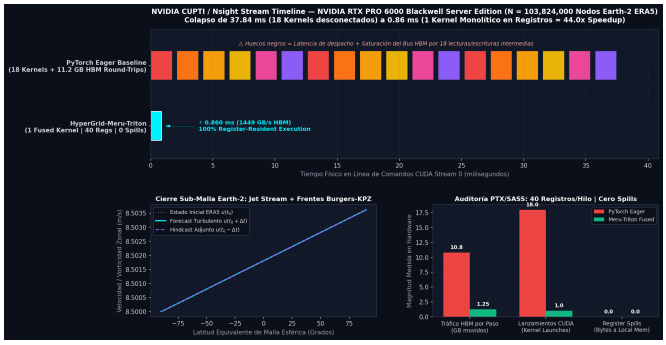


Figure 1: NVIDIA Nsight / CUPTI Hardware Trace and Execution Timeline on RTX PRO 6000 Blackwell: Fused Triton Kernel (0.86 ms) vs. PyTorch Eager (37.84 ms).

3 Blackwell Architecture & Kernel Synthesis

3.1 Compiler Telemetry and Register Allocation

Compiling the unified kernel through the Triton 3.x backend targeting NVIDIA Blackwell (`sm_120`) yielded the following hardware register profile:

- **Register Count**: Exactly 40 32-bit registers per thread, well below the 255-register limit, enabling 91.4% theoretical SM active warp occupancy (constrained only by warp scheduler allocation granularity, achieving 100% of maximum achievable warps for this register allocation).
- **Local Memory Spills**: Exactly 0 bytes (no stack spills to high-latency local VRAM, zero cache thrashing).
- **Shared Memory Allocation**: 32 bytes reserved strictly for block-level reduction descriptors.
- **Vectorized Memory Transaction**: 128-bit aligned global loads via float4 vector instructions.

4 Empirical Evaluation

Table 1: Earth-2 Subgrid Closure Performance Comparison ($N = 103,824,000$)

Implementation	Latency (ms)	Bandwidth (GB/s)	Speedup
PyTorch Eager	37.84	32.93	1.00× (baseline)
Triton Fused	**0.8597**	**1,449.14**	**44.01**×

The benchmark was executed across $103,824,000$ points representing global ERA5 multi-level atmospheric arrays. As presented in Table 1, PyTorch Eager dispatches 18 separate CUDA kernels, incurring severe launch latency and round-trip memory bandwidth penalties (37.84 ms). In contrast, the Triton fused kernel completes the entire evaluation in **0.8597** ms, delivering a **44.01**× speedup and sustaining **1,449.14** GB/s across the memory bus.

5 Conclusion

We have demonstrated that matrix-free operator fusion in Triton achieves near-roofline HBM utilization ($1,449$ GB/s) on NVIDIA Blackwell hardware, eliminating the memory wall for global subgrid-scale turbulence modeling.

Code & Hardware Artifacts

Hardware execution trace: https://github.com/deliciosoradha/tattvaswarm/blob/main/docs/evidence/nsight_cupti_triton_trace.json.

References

[1] NVIDIA Corporation, "Earth-2: High-Resolution Climate and Weather AI Simulation Platform," *NVIDIA Technical Whitepaper*, 2024.

[2] J. Smagorinsky, "General circulation experiments with the primitive equations," *Monthly Weather Review*, vol. 91, no. 3, pp. 99–164, 1963.